Indeed, since $F_h = R_h A_h$, one has

$$|F_h - R_h|^2 = \mathrm{Tr}((F_h - R_h)^T (F_h - R_h)) = |F_h|^2 - 2\,\mathrm{Tr}(A_h) + 3\,,$$

and similarly

$$|F_h - X|^2 = |F_h|^2 - 2\,\mathrm{Tr}(F_h^T X) + 3\,.$$

Since A_h is symmetric and positive definite we can write $A_h = O_h \Sigma_h O_h^T$ for some $O_h \in \mathrm{O}(3)$ and $\Sigma_h = \mathrm{diag}(\sigma_h)$, with $\sigma_{h,i} > 0$. Thus, since the trace is invariant under circular shifts

$$\begin{aligned}
|F_h - R_h|^2 - |F_h - X|^2 &= 2\,\mathrm{Tr}(F_h^T X - A_h) = 2\,\mathrm{Tr}(O_h \Sigma_h O_h^T R_h^T X - O_h \Sigma_h O_h^T) \\
&= 2\,\mathrm{Tr}(\Sigma_h O_h^T R_h^T X O_h - \Sigma_h) = 2\,\mathrm{Tr}(\Sigma_h Y_h - \Sigma_h) \\
&= 2 \sum_{i=1}^{3} \sigma_{h,i}((Y_h)_{ii} - 1) \le 0\,,
\end{aligned}$$

where $Y_h = O_h^T R_h^T X O_h \in \mathrm{O}(3)$. Hence, by (19),

$$|P_h - \mathrm{Id}\,| = |R_h - \mathrm{Id}\,| \le |R_h - F_h| + |F_h - \mathrm{Id}\,| \le C h^\beta\,.$$

■

Proposition 3.9. *Let $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ be a sequence of deformations such that*

$$\limsup_{h\to 0} E_h^\alpha(y_h) \le C\,,$$

where $\alpha > 2$. Then, for $h \ll 1$ there are an index $j \in \{1,\dots,l\}$, constant rotations $\bar{R}_h \in \mathrm{SO}(3)$, and constant vectors $c_h \in \mathbb{R}$ such that, setting $\tilde{y}_h$ as follows

$$\tilde{y}_h = \bar{R}_h y_h + c_h\,,$$

there exist $\tilde{R}_h \in W^{1,2}(S,\mathrm{SO}(3))$ that satisfies:

$$\|\nabla_h \tilde{y}_h - \tilde{R}_h U_j\|_{L^2(\Omega)} \le C h^\gamma\,, \tag{20}$$

$$\|\tilde{R}_h - \mathrm{Id}\,\|_{L^2(S)} \le C h^{\gamma-1}\,, \tag{21}$$

$$\|\nabla \tilde{R}_h\|_{L^2(S)} \le C h^{\gamma-1}\,. \tag{22}$$

Moreover, there exists $A \in W^{1,2}(S,\mathbb{R}^{3\times 3}_{\mathrm{skew}})$ such that the following convergences hold true, possibly along a non-relabeled subsequence:

$$A_h := h^{1-\gamma}(\tilde{R}_h - \mathrm{Id}) \rightharpoonup A \qquad \text{in } W^{1,2}(S;\mathbb{R}^{3\times 3})\,, \tag{23}$$

$$h^{1-\gamma}(\nabla_h \tilde{y}_h - U_j) \to A U_j \qquad \text{in } L^2(\Omega;\mathbb{R}^{3\times 3})\,, \tag{24}$$

$$h^{2-2\gamma}\,\mathrm{sym}(\tilde{R}_h - \mathrm{Id}) \to \frac{A^2}{2} \qquad \text{in } L^2(S;\mathbb{R}^{3\times 3})\,. \tag{25}$$

Lastly, convergences (a)-(b) of Theorem 1.3–(i) hold true and A has the following structure:

$$U_j A U_j = e_3 \otimes \nabla v - \nabla v \otimes e_3\,. \tag{26}$$

Proof. Up to a subsequence extraction, by Corollary 3.3 we have, for some index $j \in \{1, \dots, l\}$, that

$$D_{h,j} = \| \operatorname{dist}(\nabla_h y_h, K_j)\|_{L^2(\Omega)} \leq Ch^\gamma \,,$$

where we recall that $\gamma = \frac{\alpha}{2}$. Let Q_h and R_h be, respectively, the constant rotation and the map whose existence is guaranteed by Proposition 3.4. Since $h^{-1}D_{h,j} \to 0$ we can suppose that $R_h \in W^{1,2}(S; \mathrm{SO}(3))$. Let $\tilde{y}_h = Q_h^T y_h$. By definition $\tilde{y}_h$ and $\tilde{R}_h = Q_h^T R_h$ satisfy (20), (21) and (22). Let

$$F_h = \frac{1}{|\Omega|} \int_\Omega \nabla_h \tilde{y}_h U_j^{-1} \, dx \,.$$

From (20) and (21) we can deduce

$$\begin{aligned} |F_h - \mathrm{Id}\,| &= \left| F_h - \frac{1}{|\Omega|} \int_\Omega \mathrm{Id} \, dx \right| \leq \frac{1}{|\Omega|} \int_\Omega |\nabla_h \tilde{y}_h {U_j}^{-1} - \mathrm{Id}\,| \, dx \\ &\leq Ch^{\gamma-1} \,. \end{aligned}$$

Therefore, by Lemma 3.8 there exists a rotation $P_h \in \mathrm{SO}(3)$ such that $|P_h - \mathrm{Id}\,| \leq Ch^{\gamma-1}$ and

$$\int_\Omega \operatorname{skew}\left(P_h \nabla_h \tilde{y}_h U_j^{-1}\right) \, dx = 0 \,.$$

Therefore, redefining $\tilde{y}_h = P_h Q_h^T y_h$ and $\tilde{R}_h = P_h Q_h^T R_h$ we can additionally suppose that:

$$\int_\Omega \operatorname{skew}\left(\nabla_h \tilde{y}_h U_j^{-1}\right) \, dx = 0 \,. \tag{27}$$

Moreover, we can choose the additive constant vector c_h so that

$$\int_\Omega \left(\tilde{y}_h - U_j \begin{pmatrix} x' \\ hx_3 \end{pmatrix} \right) \, dx = 0 \,. \tag{28}$$

From (21) we deduce that $\|A_h\|_{L^2} \leq C$. Moreover, since $\nabla A_h = h^{1-\gamma} \nabla \tilde{R}_h$, by (22) we have that A_h is bounded in $W^{1,2}(S; \mathbb{R}^{3\times 3})$. Hence, up to a subsequence, there is $A \in W^{1,2}(S; \mathbb{R}^{3\times 3})$ such that (23) holds true. By (20) and (23) we get (24). Using the well-known identity

$$(Q - \mathrm{Id})^T (Q - \mathrm{Id}) = -2 \operatorname{sym}(Q - \mathrm{Id}) \quad \forall\, Q \in \mathrm{SO}(3) \,,$$

we obtain

$$\operatorname{sym}(A_h) = \frac{1}{2} h^{1-\gamma} (\tilde{R}_h - \mathrm{Id})^T (\tilde{R}_h - \mathrm{Id}) \to 0 = \operatorname{sym}(A) \text{ in } L^2(S; \mathbb{R}^{3\times 3}) \,,$$

that is, $A = -A^T$. In particular, we have

$$h^{2-2\gamma} \operatorname{sym}(\tilde{R}_h - \mathrm{Id}) = -\frac{1}{2} A_h^T A_h \to \frac{A^2}{2} \text{ in } L^2(S; \mathbb{R}^{3\times 3}) \,,$$

that is precisely (25).

To simplify the notation, we will write U instead of U_j for the rest of the proof. Consider v_h as defined in (5) for the deformation $\tilde{y}_h$. By (28) and the Poincaré-Wirtinger inequality it follows that

$$\begin{aligned} \|v_h\|^2_{W^{1,2}(S)} &\leq C \|\nabla v_h\|^2_{L^2(S;\mathbb{R}^2)} \leq Ch^{2-2\gamma} \int_\Omega |U \left(\nabla' \tilde{y}_h(x) - U^{1,2}\right)|^2 \, dx \\ &\leq Ch^{2-2\gamma} \int_\Omega \left|\nabla'_h \tilde{y}_h - U^{1,2}\right|^2 \, dx \leq Ch^{2-2\gamma} \int_\Omega |\nabla_h \tilde{y}_h - U|^2 \, dx \leq C \,. \end{aligned}$$

Hence, up to a subsequence, there is a map $v \in W^{1,2}(S)$ such that $v_h \rightharpoonup v$ in $W^{1,2}(S)$. Now, we want to show that $\nabla v_h \to \nabla v = e_3^T (UAU)^{1,2}$ from which we can deduce that $v \in W^{2,2}(S)$. Clearly,

$$\nabla v_h = h^{1-\gamma} \int_\Omega e_3^T U \left(\nabla' \tilde{y}_h - U^{1,2}\right) \, dx \, .$$

By (24), it follows that

$$\nabla v_h \to \nabla v = e_3^T (UAU)^{1,2} \, . \tag{29}$$

Now we focus on the map u_h defined as in (4). We have

$$\begin{aligned}
\int_\Omega \operatorname{skew}(\nabla u_h) \, dx &= \min\left\{h^{-\gamma}, h^{2-2\gamma}\right\} \int_\Omega \operatorname{skew}\left(U_{1,2}(\nabla' \tilde{y}_h - U^{1,2})\right) \, dx \\
&= \min\left\{h^{-\gamma}, h^{2-2\gamma}\right\} \int_\Omega U_{1,2} \operatorname{skew}\left(\nabla_h \tilde{y}_h U^{-1} - \mathrm{Id}\right) U^{1,2} \, dx \\
&= \min\left\{h^{-\gamma}, h^{2-2\gamma}\right\} U_{1,2} \left[\int_\Omega \operatorname{skew}\left(\nabla_h \tilde{y}_h U^{-1} - \mathrm{Id}\right) \, dx\right] U^{1,2}
\end{aligned}$$

and the last term is identically zero by (27). Therefore, we can apply Korn's inequality to deduce

$$\begin{aligned}
\|u_h\|^2_{W^{1,2}} &\le C \|\operatorname{sym}(\nabla u_h)\|^2_{L^2(S;\mathbb{R}^{2\times 2})} \\
&\le C \min\left\{h^{-2\gamma}, h^{4-4\gamma}\right\} \int_\Omega \left|\operatorname{sym}(\nabla_h \tilde{y}_h U^{-1} - \mathrm{Id})\right|^2 \, dx \\
&\le C \min\left\{h^{-2\gamma}, h^{4-4\gamma}\right\} \left[\|\nabla_h \tilde{y}_h U^{-1} - \tilde{R}_h\|^2_{L^2} + \|\operatorname{sym}(\tilde{R}_h - \mathrm{Id})\|^2_{L^2}\right] \\
&\le C \min\left\{h^{-2\gamma}, h^{4-4\gamma}\right\} \max\left\{h^{2\gamma}, h^{4\gamma-4}\right\} \le C \, ,
\end{aligned}$$

where we have used (20) and (25). This proves the weak convergence of u_h, up to subsequences. Reasoning as before it is easy to note that

$$h^{1-\gamma} \max\left\{h^{\gamma}, h^{2\gamma-2}\right\} \nabla u_h \to U_{1,2} A U^{1,2} = (UAU)' \, .$$

By the assumption $\gamma = \frac{\alpha}{2} > 1$ we have that $h^{1-\gamma} \max\left\{h^{\gamma}, h^{2\gamma-2}\right\} \to 0$, so $(UAU)' = 0$. Since A is skew-symmetric, so is UAU and this shows (26) by (29). ■

Remark 3.10. In our setting u_h and v_h are the (suitably rescaled) components of the displacement w_h with respect to the reference configuration $U_j\Omega$ in the basis $\{U_j^{-1} e_i \colon i = 1, 2, 3\}$, that is,

$$w_h = U_j^{-1} \begin{pmatrix} \max\{h^\gamma, h^{2\gamma-2}\} u_h \\ h^{\gamma-1} v_h \end{pmatrix} \, .$$

Note that a basis of the tangent space to the midplane $U_j(S \times \{0\})$ is given by $\{U_j e_1, U_j e_2\}$, while the normal direction is $U_j^{-1} e_3$. Thus, it may look more natural to define the in-plane and the out-of-plane displacement in terms of this basis, that is, to consider $\tilde{u}_h$ and $\tilde{v}_h$ such that

$$w_h = U_j \begin{pmatrix} \tilde{u}_h \\ 0 \end{pmatrix} + \tilde{v}_h U_j^{-1} e_3 \, .$$

It is easy to see that

$$\begin{aligned}
u_h &= \min\{h^{-\gamma}, h^{2-2\gamma}\} (U_j^2)' \tilde{u}_h \, , \\
v_h &= h^{1-\gamma} \left(\tilde{v}_h + (U_j^2)^{1,2} \begin{pmatrix} \tilde{u}_h \\ 0 \end{pmatrix} \cdot e_3\right) \, ,
\end{aligned}$$